

Here we simulate a peer-to-peer backup application using erasure coding. The source code is provided in the [storage.py](#) file. Look for the "... " patterns: those will be the parts you will have to touch to complete the first version of the simulator.

The idea is to simulate a peer-to-peer system of machines holding some data, encoding it with erasure coding, and storing the encoded blocks on peers. Peers connect and disconnect regularly, and sometimes they fail losing all the data that they have. In that case, they start downloading encoded blocks on their peers, and when they can download k of them (k is a configurable parameter) they will be able to restore all their data. The simulator allows us to simulate very long periods (say, 100 years) to verify whether data remains online. In your report, **find to what extent this is the case in the use cases you'll explore with various configurations.**

Use "`python3 storage.py --help`" to find the command-line arguments. You will see that a configuration file is required: check the [p2p.cfg](#) and [client_server.cfg](#) files as examples. They define one or more classes of machines we simulate and their characteristics. The [p2p.cfg](#) configuration specifies a system where all nodes have the same characteristics, whereas the [client_server.cfg](#) configuration is more similar to a client-server situation where some "server" machines do not own any data to back up but can store the clients' data (note that if you modify the configuration with multiple "clients", they will store some data in peer-to-peer fashion between them).

An essential thing you should keep in mind is that you should clearly define what constitutes data loss: a good definition is when there are not enough blocks anywhere in the network to recover a peer's data. This is a good metric to find out whether a given simulation has gone good or bad. Run multiple instances of the simulation to get results that have statistical significance.

Once you have completed and tested the first simulation, think about extensions or modifications to it (check the description of assignment 1 for hints about extensions) and, as with the first assignments, contact me (matteo.dellamico@unige.it) to agree on your extension. Then, as with assignment 1, write your report for it.

Last modified: Wednesday, 3 December 2025, 11:34 AM

